

EARTHQUAKE

An earthquake is a phenomenon that occurs without warning and involves violent shaking of the ground and everything over it. It results from the release of accumulated stress of the moving lithospheric or crustal plates. The earth's crust is divided into seven major plates, that are about 50 miles thick, which move slowly and continuously over the earth's interior and several minor plates. Earthquakes are tectonic in origin; that is the moving plates are responsible for the occurrence of violent shakes. The occurrence of an earthquake in a populated area may cause numerous casualties and injuries as well as extensive damage to property.

The Earthquake Risk in India

What to Do Before an Earthquake

- Identify safe places indoors and outdoors.
 - o Under a strong dining table, bed
 - o Against an inside wall
 - o Away from where glass could shatter around windows, mirrors, pictures or where heavy bookcases or other heavy furniture could fall over
 - o In the open, away from buildings, trees, telephone and electrical lines, flyovers and bridges
- Educate yourself and your family members about emergency numbers (such as those of doctors, hospitals, the police, etc) and the use of first aid and emergency kits. Learn the technique of 'Drop-Cover-Hold'.
- In case family members are separated from one another during an earthquake (a real possibility during the day when adults are at work and children are at school), develop a plan for reuniting after the disaster.
- To learn about earthquakes, participate in drills, awareness programs, and community training organized by government departments and other concerned organizations.

Constructing Earthquake-resilient House/Buildings/Infrastructure

- When doing any construction, especially houses, buildings, complexes, and other critical infrastructure, consult a structural engineer, follow BIS codes relevant to your area, and ensure the construction is earthquake-resistant. Know your seismic zone and make necessary structural changes in your house.
- Avoid flood plains and filled-up areas for construction as far as possible.
- Preserve the design and layout drawings of your house for future reference.
- Keep up with the maintenance of your house, buildings, commercial complexes, animal shelters, or any other critical infrastructure and repair them promptly. Get expert advice if there are signs of structural defects.
- Repair deep plaster cracks on walls and ceilings of the house/building regularly.
- While arranging the house furniture, fasten the shelves securely to the walls. Place large or heavy objects on lower shelves. Store breakable items such as bottled foods, glass, and china in low, closed cabinets with latches. Brace overhead light and fan fixtures.
- Hang heavy items such as pictures and mirrors away from beds, settees, and anywhere people sit.
- Secure water heaters, LPG cylinders etc., by strapping them to the walls or bolting them to the floor. Store weed killers, pesticides, and flammable products securely in closed cabinets with latches and on bottom shelves. Repair defective electrical wiring and leaky gas connections regularly. During the earthquake, it may cause potential fire risks.

What to Do During an Earthquake

- Stay as safe as possible during an earthquake. Be aware that some earthquakes are foreshocks, and a more significant earthquake might occur. Minimise your movements to a few steps that reach a safe nearby place and stay indoors until the shaking has stopped and you are sure exiting is safe.
- Help people take immediate safety actions, especially children, older people and the differently abled.
- Do not tie animals and pets

If indoors

- DROP to the ground, take cover by getting under a sturdy table or other piece of furniture, and HOLD ON until the shaking stops. If no table or desk is near you, cover your face and head with your arms and crouch in an inside corner of the building.
- Protect yourself by staying under the lintel of an inner door, in the corner of a room, under a table or even under a bed.
- Stay away from glass, windows, outside doors and walls, and anything that could fall (such as lighting fixtures or furniture).
- If you are in bed when the earthquake strikes, stay there. Hold on and protect your head with a pillow unless you are under a heavy light fixture that could fall. In that case, move to the nearest safe place.
- Use a doorway for shelter only if it is near you; if you know, it is a strongly supported, load-bearing doorway.
- Stay inside until the shaking stops and it is safe to go outside. Research has shown that most injuries occur when people inside buildings attempt to move to a different location or try to leave.
- Be aware that the electricity may go out, or the sprinkler systems or fire alarms may turn on.

If outdoors

- Do not move from where you are. However, move away from buildings, trees, streetlights, and utility wires.
- If you are in an open space, stay there until the shaking stops. The greatest danger exists directly outside buildings, at exits, and alongside exterior walls. Most earthquake-related casualties result from collapsing walls, flying glass, and falling objects.

If in a moving vehicle

- Stop as quickly as safety permits and stay in the vehicle. Avoid stopping near or under buildings, trees, overpasses, and utility wires.
- Once the earthquake has stopped, proceed cautiously. Avoid roads, bridges, or ramps that the earthquake might have damaged.

If trapped under debris

- Do not light a match.

- Do not move about or kick up dust.
- Cover your mouth with a handkerchief or clothing.
- Tap on a pipe or wall so rescuers can locate you. Use a whistle if one is available.
- Shout only as a last resort. This will help you conserve energy, and shouting can cause you to inhale dangerous amounts of dust.

After the Earthquake

- Do not enter damaged buildings.
- Use stairs and not lifts or elevators. Move cautiously, and check for unstable objects and other hazards above and around you. Check yourself for injuries.
- Anticipate aftershocks, especially after a major earthquake.
- Stay away from buildings, electric poles, trees or any other hazardous places till the safety of these structures is technically assessed/audited and approved by concerned government authorities.
- Do not spread and/ or believe in rumours or misinformation. Follow instructions and warnings from concerned government agencies and cooperate with them to conduct search and rescue operations.
- Leave a message stating where you are going if you must evacuate your house.
- Do not drive around the damaged areas, as rescue and relief operations need roads for mobility.
- Do not attempt to cross bridges/flyovers, which may have been damaged.

How to protect your property

Brace or anchor high or top-heavy objects. During an earthquake, these items can fall over, causing damage or injury.

Secure items that might fall (televisions, books, computers, etc.). Falling items can cause damage or injury.

Install strong latches or bolts on cabinets. The contents of cabinets can shift during the shaking of an earthquake. Latches will prevent cabinets from flying open and contents from falling out.

Move large or heavy objects and fragile items (glass or china) to lower shelves. There will be less damage and less chance of injury if these items are on lower shelves.

Store breakable items such as bottled foods, glass, and china in low, closed cabinets with latches. Latches will help keep contents of cabinets inside.

Store weed killers, pesticides, and flammable products securely in closed cabinets with latches, on bottom shelves. Chemical products will be less likely to create hazardous situations from lower, confined locations.

Hang heavy items, such as pictures and mirrors, away from beds, couches, and anywhere people sit. Earthquakes can knock things off walls, causing damage or injury.

Brace overhead light fixtures. During earthquakes, overhead light fixtures are the most common items to fall, causing damage or injury.

Strap the water heater to wall studs. The water heater may be your best source of drinkable water following an earthquake. Protect it from damage and leaks.

Bolt down any gas appliances. After an earthquake, broken gas lines frequently create fire hazards.

Install flexible pipe fittings to avoid gas or water leaks. Flexible fittings will be less likely to break.

Repair any deep cracks in ceilings or foundations. Get expert advice if there are signs of structural defects. Earthquakes can turn cracks into ruptures and make smaller problems bigger.

Check to see if your house is bolted to its foundation. Homes bolted to their foundations are less likely to be severely damaged during earthquakes. Homes that are not bolted have been known to slide off their foundations, and many have been destroyed because they are uninhabitable.

Consider having your building evaluated by a professional structural design engineer. Ask about home repair and strengthening tips for exterior features, such as porches, front and back decks, sliding glass doors, canopies, carports, and garage doors. Learn about additional ways you can protect your home. A professional can give you advice on how to reduce potential damage.

Emergency Kit

Battery operated torch

Extra batteries

Battery operated radio

First aid kit and manual

Emergency food (dry items) and water (packed and sealed)

Candles and matches in a waterproof container

Knife

Chlorine tablets or powdered water purifiers

Can opener.

Essential medicines

Cash, Aadhar Card and Ration Card

Thick ropes and cords

Sturdy shoes